ENGINEERING ECONOMICS

LECTURE - 07

Simple and Compound Interest

Interest Periods Interest is usually charged on amount for a period of one year. Interest rates are quoted for periods other than one year, known as interest periods.

Simple Interest: Under simple interest, the interest owed upon repayment of a loan is proportional to the length of time the principal sum has been borrowed.

The interest earned may be found in the following manner. Let **I** represents the interest earned, **P** the principal amount, **n** the number of interest periods, and **i** the interest rate. Then,

$$\mathbf{I = Pni}$$

Suppose that Rs1000 is borrowed at a simple interest rate of 12% per annum. At the end of one year, the interest owed would be

$$\underline{\mathbf{I = 1000(1)(0.12) = Rs120}}$$

The principal plus the interest would be Rs1120 and would be due at the end of one year.

A simple-interest loan can be made for any period of time.

Interest and principal become due only at the end of the time period.

When it is necessary to calculate the interest due for a fraction of year, it is common to represent that fraction as the ratio of the number of days in the loan to the total days in a year.

For example, on a loan of Rs1000 at an interest rate of 12% per annum, for a period March 1 to May 20, the interest due on May 20 along with the principal sum of Rs1000 would be

<u>0.12(1000) (81 / 365) = Rs26.63</u>

When a loan is made for several interest periods, interest is calculated and payable at the end of each interest period.

There are a number of loan repayment plans. These range from ***paying the interest when it is due*** to accumulating the interest ***until the loan is due***.

For example, the payments on a 4-year loan of Rs1000 at 16% interest per annum, payable when due, would be calculated as shown.

Yr	Amount owed at beginning of yr Rs	Interest to be paid at end of yr Rs	Amount owed at end of yr Rs	Amount to be paid by the borrower at end of yr-Rs
1	1000	160	1160	160
2	1000	160	1160	160
3	1000	160	1160	160
4	1000	160	1160	1160

*If the borrower does not pay the interest earned at the end of each period and is charged interest on the total amount owed (principal plus interest), the interest is said to be **compounded**.*

The interest owed in the previous year becomes part of the total amount owed for this year. This year's interest charge includes interest that has been earned on previous interest charges.

For example, a loan of Rs 1000 at 16% interest compounded annually for a 4-year period will produce the results shown.

Year	Amount (Rs) owed at beginning of year (A)	Interest (Rs) to be added to loan at end of year (B)	Amount (Rs) owed at end of year (A+B)	Amount (Rs) paid by borrower at end of year
1	1000.00	1000 X 0.16 =160.00	1000(1.16)= 1160	00.00
2	1160.00	1160 x 0.16 =185.60	$1000(1.16)^2$= 1345.60	00.00
3	1345.60	1345x 0.16 =215.30	$1000(1.16)^3$= 1560.90	00.00
4	1560.90	1560.90x 0.16 =249.75	$1000(1.16)^4$= 1810.64	1810.64

Describing Cash Flows Over Time

Cash flow is the actual inflow (receipts) and outflow (disbursements) at different points in time that occur over the life of an investment.

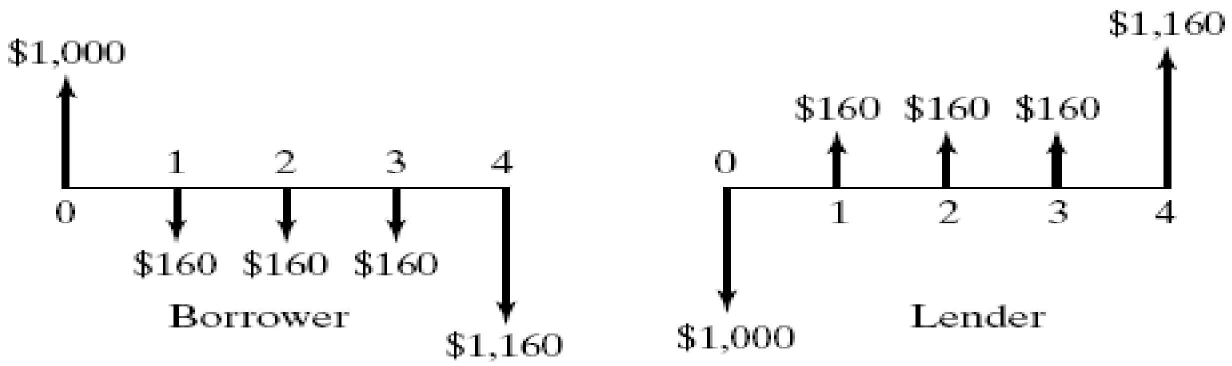
Cash flow diagram will provide the information necessary for analyzing an investment proposal.

A cash flow diagram represents receipts received during a period of time by an upward arrow (an increase in cash) located at the period’s end.

The arrow’s height may be proportional to the magnitude of the receipts during that period.

Similarly, disbursements during a period are represented by a downward arrow (a decrease in cash).

These arrows are then placed on a time scale that spans all time periods covered by the proposed investment.



Note: Since there are two parties to every transaction, it is important to note that the cash flow directions in cash flow diagrams depend upon the point of view taken.

Net Cash Flow

When an investment alternative has both receipts and disbursements occurring simultaneously, a net cash flow may be calculated.

Net cash flow is the arithmetic sum of the receipts (+) and the disbursements (-) that occur at the same point in time.

Compounding Frequency Considerations

The loan agreements may require that the interest be paid more frequently, such as each half year, each quarter or each month.

Such agreements result in interest periods of one half-year, one quarter year, or one twelfth year, and the compounding of interest twice, four times, or twelve times a year, respectively.

- **Every year – once a year (at the end):**
  - **(Annually)**
- **Every 6 months – 2 times a year:**
  - **(Semi-annually)**
- **Every quarter – 4 times a year:**
  - **(Quarterly)**
- **Every month – 12 times a year:**
  - **(Monthly)**
- **Every day – 365 times a year:**
  - **(Daily)**
- **Continuous – infinite number of compounding periods in a year!**

Quotation of Interest Rates

- Interest rates can be quoted in several different ways
- Examples:
  - **12% per year**
  - **1% per month**
  - **12% per year, compounded monthly**
- Thus, you have to know the various ways in order to determine:
  - **Which interest rates are equivalent**
  - **Which interest rates are the best**

Discussion